MUGIL MURALEEDHARAN

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Full Stack Engineer specializing in scalable distributed systems and real-time infrastructure. Sole architect of a production-grade social platform supporting **10K+ concurrent users** and **100K+ daily API** requests with **<200ms p95 latency** and 99.9% uptime. Strong in backend architecture, system design, real-time messaging, streaming systems, and cloud-native deployment. Additionally built production **ELT/ETL pipelines** with Airflow, dbt, Snowflake, and Kafka, processing **100K+ records** daily with sub-100ms streaming latency

TECHNICAL SKILLS

- **Languages:** Python, JavaScript/TypeScript, SQL, HTML5, CSS3
- **Frontend Technologies:** React.js, Vue.js, AngularJS, HTML5, CSS3, JavaScript (ES6+), TypeScript, Responsive Design, Progressive Web Apps, State Management
- **Backend & APIs:** FastAPI, Node.js (Express), RESTful APIs, GraphQL, WebSockets, Microservices, Auth (JWT, OAuth2), Rate Limiting,
- **Database & Caching:** PostgreSQL, MySQL, MongoDB, SQLite, Redis (caching, pub/sub), Distributed Systems.
- **Cloud & DevOps:** AWS (EC2, S3), Cloudflare (R2, CDN), Docker, Nginx, CI/CD (GitHub Actions)
- **Mobile Development:** Flutter (cross-platform), Android (Kotlin), iOS (Swift)
- **Video & Streaming:** HLS/DASH, FFmpeg, Adaptive Streaming, ExoPlayer, AVPlayer, CDN integration

PROFESSIONAL EXPERIENCE

Founding Full Stack Engineer - Socionn - https://www.socionn.com

Sole architect and engineer of a real-time social media platform across web, Android, iOS, backend, and infrastructure. Owned the entire product lifecycle from system design to production operations.

- Architected an end-to-end distributed platform supporting **10K+ concurrent users** and **100K+ daily API requests** with **<200ms p95 latency**, designing service boundaries across feeds, chat, notifications, media, and authentication domains.
- Modeled and optimized **25+ table PostgreSQL schema** for social graph queries and feed generation at scale.
- Reduced database load by **70%** via multi-layer caching (Redis + in-process LRU).
- Built WebSocket-based messaging system supporting **5K+ concurrent sessions** with delivery guarantees and **<50ms** notification latency.
- Engineered async video processing pipeline (FFmpeg + HLS/DASH) handling **500+ uploads/day**, reducing **buffering by 60%.**
- Containerized services with Docker and implemented **CI/CD pipelines** achieving **99.9% uptime** with zero-downtime deployments.
- Implemented authentication, **OAuth2 flows**, **rate limiting,** and abuse protection passing security review with zero critical vulnerabilities.

AI-POWERED MAIL CLIENT WITH AUTONOMOUS UI CONTROL VIA LLM FUNCTION CALLING – Live Demo

Next.js (TypeScript) · FastAPI · Gmail API · OAuth2 · SSE · Zustand · OpenAI Function Calling

- Designed **deterministic LLM function-calling pipeline** enabling natural language → typed UI state transitions.
- Built **real-time inbox sync** using **Server-Sent Events**, Gmail **History API**, and resilient polling fallback.
- Engineered **RFC-compliant email threading**, MIME parsing, and token auto-refresh for long-lived sessions.
- Implemented **finite state machine + action dispatcher** for single-cycle command execution and idempotent updates.
- Implemented **OAuth2 authorization flow**, secure token storage, and refresh rotation.
- Optimized **client state management** with **Zustand** to prevent redundant renders and race conditions.

BACKEND & INFRASTRUCTURE – Live Demo

E-Commerce Intelligence Analytics System | *Python, Airflow, AWS S3, Snowflake, dbt, Docker, Power BI*

- Architected fully automated ELT pipeline processing 50K+ daily transactions from AWS S3 to Snowflake, reducing manual processing by 90%.
- Designed 3-layer dbt transformation architecture (staging → intermediate → marts) with 25+ models implementing star schema for analytics.
- Orchestrated workflows using Apache Airflow with 12+ DAGs, implementing task dependencies, retry logic, and SLA monitoring.

Live Air Traffic Monitoring Pipeline | *Apache Kafka, Python, FastAPI, PostgreSQL, Docker*

- Engineered high-throughput Kafka pipeline processing 10K+ flight telemetry records per minute with sub-100ms latency.
- Designed Kafka topic architecture with 8-partition strategy achieving parallel processing and 300% throughput improvement.
- Implemented real-time data enrichment (geocoding, airline mapping) using Python consumer groups, transforming raw data to analytics-ready format.

Land Deed Extraction Pipeline (OCR + LLM) | *Python, FastAPI, Tesseract OCR, GPT-4 API, PostgreSQL*

- Designed multi-stage automated pipeline (ingestion → OCR → cleaning → LLM parsing → validation) processing 500+ documents daily, reducing manual entry by 85%.
- Leveraged LLM APIs with engineered prompts to extract structured fields from unstructured OCR text with 92% field-level accuracy.

KEY ACHIEVEMENTS

10K+ Concurrent Users — Sole-architected production distributed platform with <200ms p95 latency

70% DB Load Reduction — Multi-layer caching (Redis + LRU), enabling 3x traffic spike tolerance

500+ Video Uploads/Day — Async FFmpeg pipeline with HLS/DASH, reducing buffering by 60%

90% Manual Processing Cut — ELT pipeline processing 50K+ daily transactions (S3 → Snowflake)

300% Throughput Boost — Kafka 8-partition architecture for 10K+ flight records/min

99.9% Uptime — Maintained SLA across 6 months of continuous production operation

100K+ Daily API Requests — Handled at scale across feeds, chat, notifications, media & auth

5K+ Concurrent Chat Sessions — WebSocket messaging with <50ms delivery latency & guarantees

92% Field-Level Accuracy — OCR + LLM extraction pipeline processing 500+ documents/day

Zero Critical Vulnerabilities — Auth, OAuth2, rate limiting & abuse prevention passed security review

EDUCATION

MSc in Big Data Analytics | St. Joseph's University | Bengaluru, India | 2024 – 2026
BSc in Computer Science | St. Joseph's College | Bengaluru, India | 2021